



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to MRN TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

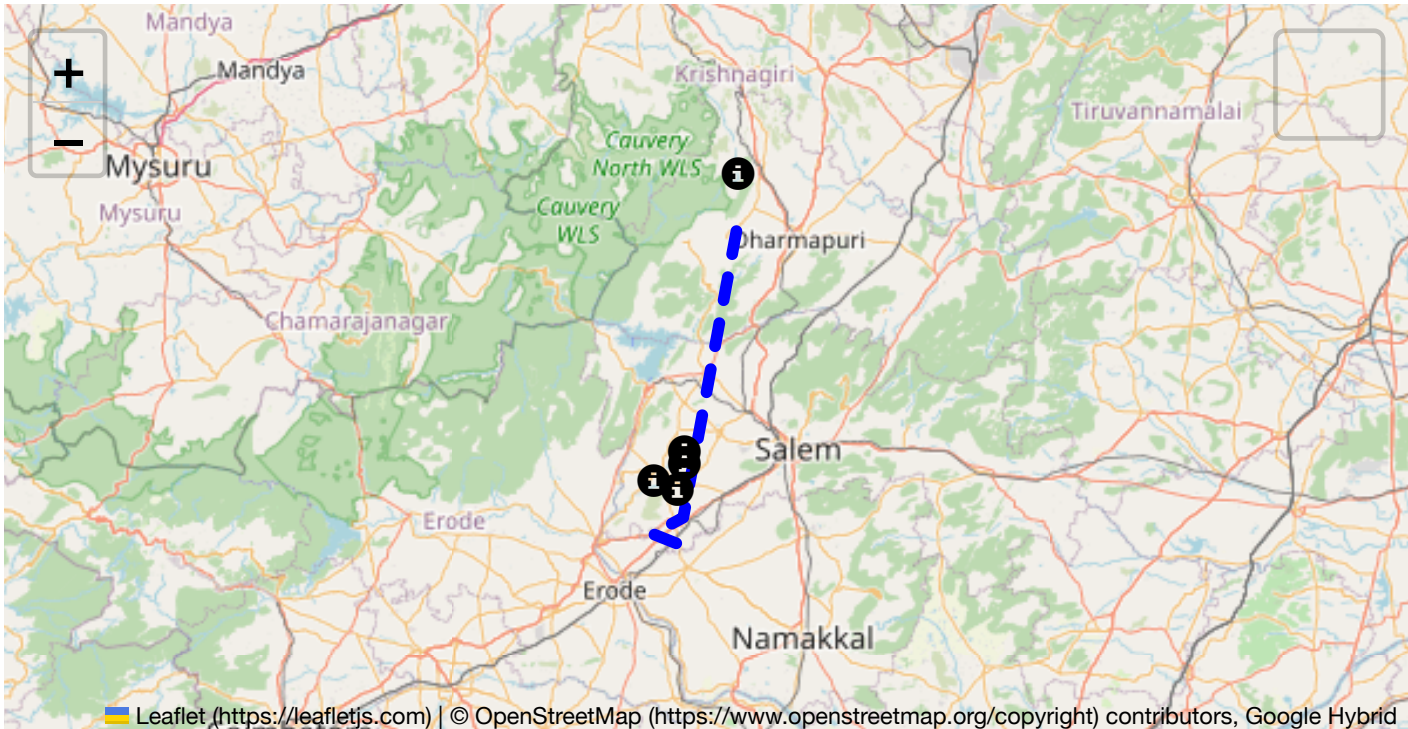
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

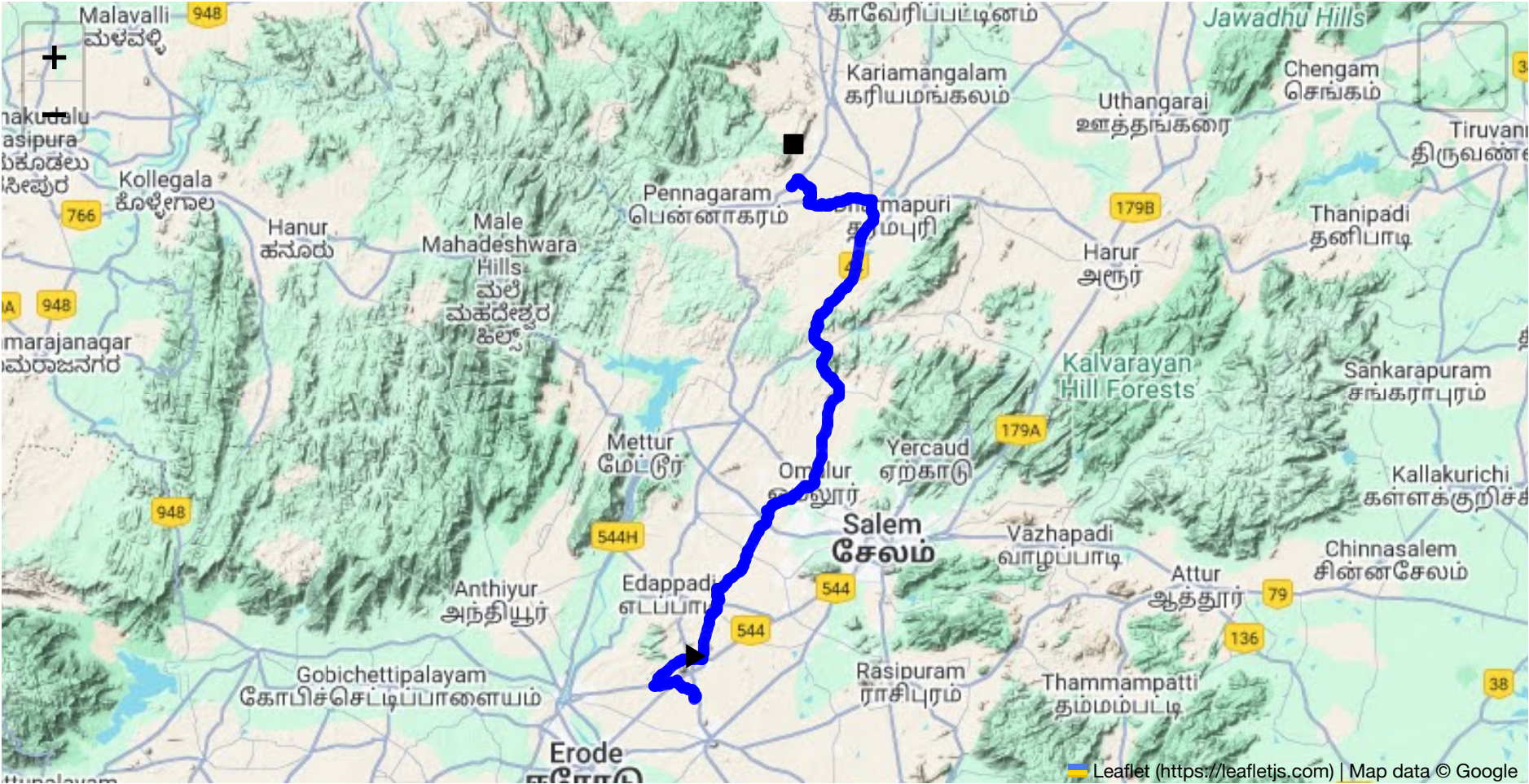
Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 122.13 km
Estimated Duration: 2.7 hours
Adjusted Duration (Heavy Vehicle): 3.3 hours

Start: (11.4381, 77.8734)
End: (12.160818, 78.014197)



Welcome to the Journey Risk Management Study

Route Overview

The route from CVQF+23W, Sangagiri to 5267+6FV, Indur is approximately 122.13 kilometers and mostly rural, passing through several towns such as Padaiveedu, Ayveli, and Olakkachinnanur. Key roads include Bhavani Main Road and State Highway 86.

Typical Weather Conditions and Hazards

Tamil Nadu generally experiences a tropical climate with three main seasons: summer (March to June), monsoon (July to September), and winter (October to February). During the monsoon, heavy rainfall can lead to flooding, affecting visibility and road conditions. Heat during summer may cause road surface damage.

Traffic Patterns

Traffic is generally moderate but can become heavy near towns and junctions during peak hours, typically from 8:00 AM to 10:30 AM and 5:00 PM to 8:00 PM. Areas near markets and town centers may experience congestion due to local activities.

Road Quality and Infrastructure

While the route primarily consists of state highways, which are generally maintained, some segments, especially rural connections, may have uneven surfaces, potholes, or lack proper lane markings. Signage can be sparse in certain areas.

Suggestions for Alternative Routes

In emergencies, consider alternate state highways that run parallel to the main route. Local roads may provide bypasses around congested towns but check their condition periodically.

Local Regulations on Hazardous Materials

Transport of hazardous materials is subject to state regulations, including restrictions on timing and routes. Permits are mandatory, and vehicles should be clearly marked. Night-time transport may require special permissions.

Historical Incidents

While specific incidents about hazardous materials might not be frequently reported, heavy vehicle accidents have occurred on the state highways due to sharp turns and poor visibility during heavy rains.

Environmental Considerations

The area has agricultural lands, requiring careful handling of hazardous materials to prevent contamination. Proximity to water bodies along the route necessitates caution to avoid spillages.

Communication Coverage

Coverage is generally good along highways and near towns. However, some rural stretches may experience weak signals or dead zones, particularly in low-lying areas or dense landscape.

Estimated Emergency Response Times

Emergency response in rural areas may range from 30 to 60 minutes, depending on proximity to towns with fire stations or healthcare facilities.

Overall Risk Assessment

- **Weather:** Medium risk due to seasonal variations.
- **Traffic and Congestion:** Moderate in town centers; significant during peak hours.
- **Road Quality:** Variable, requiring attention to road conditions.
- **HazMat Regulations:** Stringent regulations require adherence.
- **Emergency Services:** Limited quick response outside urban areas.
- **Environmental Impact:** High risk near agricultural lands and water bodies.

Conclusion

The route requires careful planning, particularly during adverse weather conditions or peak traffic times. Drivers should be equipped with emergency contacts and communication devices, and adhere strictly to local regulations regarding hazardous materials. Regular updates on road conditions and local advisories can further enhance safety on this route.

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45353, 77.85697	15 KM/Hr
4	Turn	Medium	11.45781, 77.85203	30 KM/Hr
5	Turn	High	11.46307, 77.84941	15 KM/Hr
6	Turn	High	11.45542, 77.81725	15 KM/Hr
7	Turn	High	11.45605, 77.81683	15 KM/Hr
8	Turn	Medium	11.49383, 77.88501	30 KM/Hr
9	Blind Spot	Blind Spot	11.49400, 77.88538	10 KM/Hr
10	Turn	High	11.49469, 77.88488	15 KM/Hr
11	Turn	High	11.55614, 77.89764	15 KM/Hr
12	Turn	Medium	11.55680, 77.89969	30 KM/Hr
13	Turn	High	11.58218, 77.90845	15 KM/Hr
14	Turn	High	11.58330, 77.90642	15 KM/Hr
15	Turn	Medium	11.59189, 77.90645	30 KM/Hr
16	Turn	Medium	11.59328, 77.90809	30 KM/Hr
17	Turn	Medium	11.63867, 77.94841	30 KM/Hr
18	Turn	High	11.64217, 77.95200	15 KM/Hr
19	Turn	Medium	11.64712, 77.95118	30 KM/Hr
20	Turn	Medium	11.67858, 77.96834	30 KM/Hr
21	Turn	Medium	11.70062, 77.97961	30 KM/Hr
22	Turn	High	11.70218, 77.97866	15 KM/Hr
23	Turn	Medium	11.71399, 77.99580	30 KM/Hr
24	Turn	Medium	11.71917, 78.01123	30 KM/Hr
25	Turn	Medium	11.73769, 78.03451	30 KM/Hr
26	Turn	Medium	11.73785, 78.03551	30 KM/Hr
27	Turn	High	11.73651, 78.03871	15 KM/Hr
28	Turn	Medium	11.74271, 78.04717	30 KM/Hr
29	Turn	Medium	11.86847, 78.08367	30 KM/Hr
30	Turn	Medium	11.87942, 78.08197	30 KM/Hr
31	Turn	Medium	11.92585, 78.06507	30 KM/Hr
32	Turn	Medium	11.93704, 78.05402	30 KM/Hr
33	Turn	High	11.95818, 78.04978	15 KM/Hr
34	Turn	Medium	11.95988, 78.04948	30 KM/Hr
35	Turn	Medium	11.96304, 78.05136	30 KM/Hr
36	Turn	High	11.96399, 78.05135	15 KM/Hr
37	Turn	High	11.96580, 78.04923	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
38	Turn	Medium	11.96675, 78.04939	30 KM/Hr
39	Turn	Medium	11.96791, 78.05070	30 KM/Hr
40	Turn	Medium	12.12058, 78.13283	30 KM/Hr
41	Turn	High	12.14089, 78.12817	15 KM/Hr
42	Turn	Medium	12.14449, 78.09686	30 KM/Hr
43	Turn	Medium	12.13370, 78.06372	30 KM/Hr
44	Turn	High	12.13572, 78.05870	15 KM/Hr
45	Turn	Medium	12.13532, 78.05787	30 KM/Hr
46	Turn	Medium	12.13448, 78.05740	30 KM/Hr
47	Blind Spot	Blind Spot	12.13654, 78.03928	10 KM/Hr
48	Turn	Medium	12.13945, 78.04029	30 KM/Hr
49	Turn	High	12.14135, 78.03988	15 KM/Hr
50	Turn	High	12.14158, 78.03948	15 KM/Hr
51	Turn	Medium	12.14481, 78.04006	30 KM/Hr
52	Turn	Medium	12.14650, 78.04137	30 KM/Hr
53	Turn	High	12.15016, 78.04240	15 KM/Hr
54	Turn	High	12.15189, 78.04012	15 KM/Hr
55	Turn	Medium	12.15323, 78.04034	30 KM/Hr
56	Blind Spot	Blind Spot	12.15415, 78.03930	10 KM/Hr
57	Turn	Medium	12.15462, 78.03972	30 KM/Hr
58	Turn	High	12.15750, 78.04027	15 KM/Hr
59	Turn	High	12.17086, 78.02442	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
6	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
7	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
8	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
12	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
13	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
14	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
15	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
16	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
17	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
18	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
19	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
20	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
21	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
22	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
24	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
25	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
26	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
27	hospital	Government Hospital, Indur	12.1326826, 78.0577248	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
9	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
23	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45353, 77.85697



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45781, 77.85203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46307, 77.84941



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45605, 77.81683



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49383, 77.88501



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49400, 77.88538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 77.88488



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.55614, 77.89764



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58330, 77.90642



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59189, 77.90645



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59328, 77.90809



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63867, 77.94841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.64217, 77.95200



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64712, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67858, 77.96834



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70062, 77.97961



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70218, 77.97866



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71399, 77.99580



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71917, 78.01123



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73769, 78.03451



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73785, 78.03551



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.73651, 78.03871



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.74271, 78.04717



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.86847, 78.08367



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.87942, 78.08197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92585, 78.06507



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93704, 78.05402



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95818, 78.04978



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95988, 78.04948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96304, 78.05136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96399, 78.05135



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96580, 78.04923



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96675, 78.04939



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96791, 78.05070



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.12058, 78.13283



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14089, 78.12817



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14449, 78.09686



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.13370, 78.06372



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.13572, 78.05870



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.13532, 78.05787



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.13448, 78.05740



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.13654, 78.03928



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.13945, 78.04029



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14135, 78.03988



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.14158, 78.03948



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14481, 78.04006



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.14650, 78.04137



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.15016, 78.04240



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.15189, 78.04012



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.15323, 78.04034



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 12.15415, 78.03930



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.15462, 78.03972



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.15750, 78.04027



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.17086, 78.02442

